

Multiple Choice (10 marks)

1. Cellular uptake of cholesterol occurs through internalization of low density lipoprotein-cholesterol complexes. Cholesterol uptake involves all of the following EXCEPT
 - (a) cell-surface receptors
 - (b) adhesion plaques
 - (c) endosome formation
 - (d) receptor-mediated endocytosis
2. The target of digestion of a nucleosome dimer to nucleosome monomers by DNase is
 - (a) histones H₂A, H₂B, H₃, and H₄
 - (b) linker DNA
 - (c) histone H₂A and H₂B only
 - (d) the nucleosome linker
 - (e) histone H₃ only
3. What are the different types of point mutations?
 - (a) Frame shift mutations and nucleotide repeats
 - (b) Base substitutions and small insertions and deletions
 - (c) Base additions and large insertions
 - (d) Silent mutations and missense mutations
 - (e) Duplication mutations and triplet repeat expansions
4. What factors affect the rate of transpiration?
 - (a) Temperature, humidity, and air currents
 - (b) Plant species, root depth, and wind speed
 - (c) Soil fertility, irrigation methods, and cloud cover
 - (d) Water salinity, leaf color, and time of day
 - (e) Leaf surface area, stem length, and moon phase
5. In what way does Lamarck's theory of acquired characteristics not agree with present evidence?
 - (a) Acquired characteristics are inherited through a blending of parental traits
 - (b) Acquired characteristics are inherited but do not influence evolution
 - (c) Acquired characteristics are inherited randomly, regardless of their use
 - (d) Acquired characteristics cannot be inherited
 - (e) Acquired characteristics are inherited based on use and disuse

True / False (10 marks)

Answer True or False

6. True or False: The quantity of DNA in a eukaryotic cell typically doubles during telophase of the cell cycle.
7. True or False: Proprioceptors provide the body with information about its position and movement, enabling actions without visual input.
8. True or False: A cell is the fundamental organizational unit of life.
9. True or False: The glomerular filtration rate is mainly influenced by the concentration of sodium in the distal tubule and renal blood flow.
10. True or False: Autotrophs evolved from heterotrophs directly due to the overabundance of organic nutrients in the environment.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the evidence supporting the transmission of impulses in plants, focusing on examples such as the sensitive plant, *Mimosa pudica*, and their physiological responses to stimuli.
12. Discuss the role of fermentation in cellular respiration when oxygen is scarce, emphasizing how it regenerates NAD^+ and its implications for energy production.

End of Paper